

Common TCP Ports Reference

Introduction

TCP (Transmission Control Protocol) ports enable communication between applications over a network. During network reconnaissance and security assessments, identifying open ports helps determine which services are running on a target system and whether they present potential security risks. This document provides a reference for commonly used TCP ports, their associated services, and recommended security practices.

Common TCP Ports

Port	Service	Protocol	Description	Common Security Risks	Recommendation
20/21	FTP	TCP	File Transfer Protocol used for transferring files.	Plaintext credentials, unauthorized file access.	Use SFTP or FTPS instead of FTP.
22	SSH	TCP	Secure remote administration.	Brute-force attacks, weak passwords.	Use strong passwords, SSH keys, and disable root login.
23	Telnet	TCP	Remote terminal access.	Sends all data in plaintext.	Disable Telnet and use SSH instead.
25	SMTP	TCP	Sends email between mail servers.	Open relay abuse, spam.	Configure authentication and enable encryption.
53	DNS	TCP/UDP	Resolves domain names into IP addresses.	DNS amplification and cache poisoning attacks.	Restrict recursion and keep DNS software updated.
80	HTTP	TCP	Unencrypted web traffic.	Data interception, session hijacking.	Redirect users to HTTPS.
110	POP3	TCP	Retrieves emails from a mail server.	Plaintext authentication.	Use POP3S or IMAPS where possible.
139	NetBIOS	TCP	Windows file and printer sharing.	Information disclosure and unauthorized access.	Disable if not required and restrict access.
143	IMAP	TCP	Email retrieval and synchronization.	Plaintext communication if encryption is absent.	Use IMAPS (Port 993).
161	SNMP	UDP	Network device monitoring and management.	Default community strings can expose device information.	Change default credentials and restrict access.
389	LDAP	TCP	Directory services authentication.	Information disclosure.	Use LDAPS (Port 636).
443	HTTPS	TCP	Secure encrypted web communication.	Misconfigured certificates, outdated TLS versions.	Use valid certificates and modern TLS versions.
445	SMB	TCP	Windows file and printer sharing.	SMB exploits, ransomware propagation.	Disable SMBv1 and restrict access.
3306	MySQL	TCP	MySQL database service.	Unauthorized database access.	Restrict remote connections and use strong credentials.

Port	Service	Protocol	Description	Common Security Risks	Recommendation
3389	RDP	TCP	Remote Desktop access for Windows.	Brute-force attacks and unauthorized access.	Enable MFA, use VPN, and restrict access.
5432	PostgreSQL	TCP	PostgreSQL database service.	Database exposure.	Limit network access and use authentication.
5900	VNC	TCP	Remote desktop sharing.	Weak authentication.	Use strong passwords and encrypt traffic.
8080	HTTP Alternate	TCP	Alternative web server port.	Exposed web applications.	Secure web applications and monitor access.

Importance of Port Scanning

Port scanning helps security professionals:

- Discover active hosts on a network.
 - Identify running network services.
 - Detect unnecessary exposed services.
 - Assess the network attack surface.
 - Verify firewall configurations.
 - Support vulnerability assessments and security audits.
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Security Best Practices

- Close unused ports.
 - Enable and properly configure firewalls.
 - Disable unnecessary network services.
 - Regularly perform vulnerability assessments.
 - Monitor network traffic using Wireshark.
 - Keep operating systems and applications updated.
 - Use strong authentication and multi-factor authentication (MFA).
 - Replace insecure protocols (FTP, Telnet, HTTP) with secure alternatives (SFTP, SSH, HTTPS).
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Conclusion

Understanding common TCP ports is essential for network administrators and cybersecurity professionals. Proper management of network services and regular security assessments reduce the attack surface and improve the overall security posture of an organization.